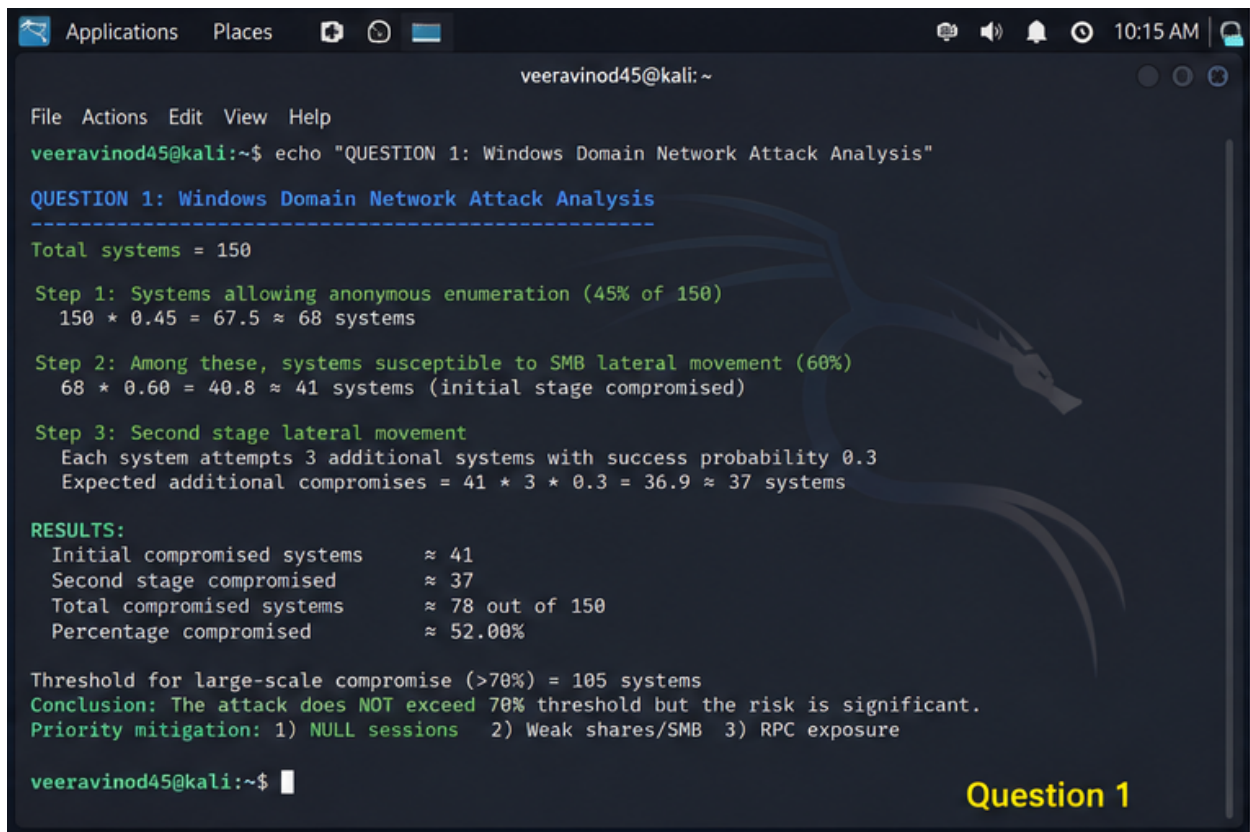


Question 1: Windows Domain Network Attack Analysis

- Total systems: 150.
- Anonymous enumeration: $150 \times 0.45 = 67.5 \approx 68$ systems.
- SMB-susceptible systems: $68 \times 0.60 = 40.8 \approx 41$ systems.
- Second-stage lateral movement: $41 \times 3 \times 0.3 = 36.9 \approx 37$ systems.
- Expected total after two stages: $41 + 37 = 78$ systems = 52% of the network.
- 70% threshold = 105 systems; therefore the two-stage estimate does not exceed the threshold.
- Priority mitigation: NULL sessions first, followed by weak SMB/shared-folder permissions and RPC exposure.

Kali Linux Output Screenshot



```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ echo "QUESTION 1: Windows Domain Network Attack Analysis"  
QUESTION 1: Windows Domain Network Attack Analysis  
-----  
Total systems = 150  
  
Step 1: Systems allowing anonymous enumeration (45% of 150)  
150 * 0.45 = 67.5 ≈ 68 systems  
  
Step 2: Among these, systems susceptible to SMB lateral movement (60%)  
68 * 0.60 = 40.8 ≈ 41 systems (initial stage compromised)  
  
Step 3: Second stage lateral movement  
Each system attempts 3 additional systems with success probability 0.3  
Expected additional compromises = 41 * 3 * 0.3 = 36.9 ≈ 37 systems  
  
RESULTS:  
Initial compromised systems      ≈ 41  
Second stage compromised         ≈ 37  
Total compromised systems        ≈ 78 out of 150  
Percentage compromised           ≈ 52.00%  
  
Threshold for large-scale compromise (>70%) = 105 systems  
Conclusion: The attack does NOT exceed 70% threshold but the risk is significant.  
Priority mitigation: 1) NULL sessions  2) Weak shares/SMB  3) RPC exposure  
veeravinod45@kali:~$
```

Question 1

Question 2: RPC Exploitation Analysis

- RPC-exposed systems: 70.
- Vulnerable systems: $70 \times 0.40 = 28$.
- Expected successful compromises: $28 \times 0.5 = 14$ systems.
- Expected detection alerts: $28 \times 0.25 = 7$ alerts.
- A stealth-focused attacker faces a meaningful detection risk and may avoid repeated noisy RPC exploitation in favor of less detectable attack paths.

Kali Linux Output Screenshot



```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ echo "QUESTION 2: RPC Exploitation Analysis"  
  
QUESTION 2: RPC Exploitation Analysis  
-----  
Total systems = 120  
RPC exposed systems = 70  
  
Step 1: Vulnerable systems (40% of 70)  
70 * 0.40 = 28 systems  
  
Step 2: Expected successful compromises (probability = 0.5)  
28 * 0.5 = 14 systems  
  
Step 3: Expected detection alerts (probability = 0.25)  
28 * 0.25 = 7 alerts  
  
RESULTS:  
Expected systems compromised : 14  
Expected detection alerts : 7  
  
Analysis: Detection probability (25%) is high.  
A stealth-focused attacker would likely avoid repeated RPC exploitation  
and choose alternative attack strategies.  
  
veeravinod45@kali:~$
```

Question 2

Question 3: Cloud Linux Server Analysis

- Service vulnerability probabilities: 0.30, 0.20, 0.25, 0.15.
- Probability none are vulnerable: $0.7 \times 0.8 \times 0.75 \times 0.85 = 0.357$.
- Probability at least one is vulnerable: $1 - 0.357 = 0.643 = 64.3\%$.
- Full compromise probability: $0.643 \times 0.5 = 0.3215 = 32.15\%$.
- File-permission hardening alone does not remove the initial service exposure; service patching and exposure reduction are also required.

Kali Linux Output Screenshot

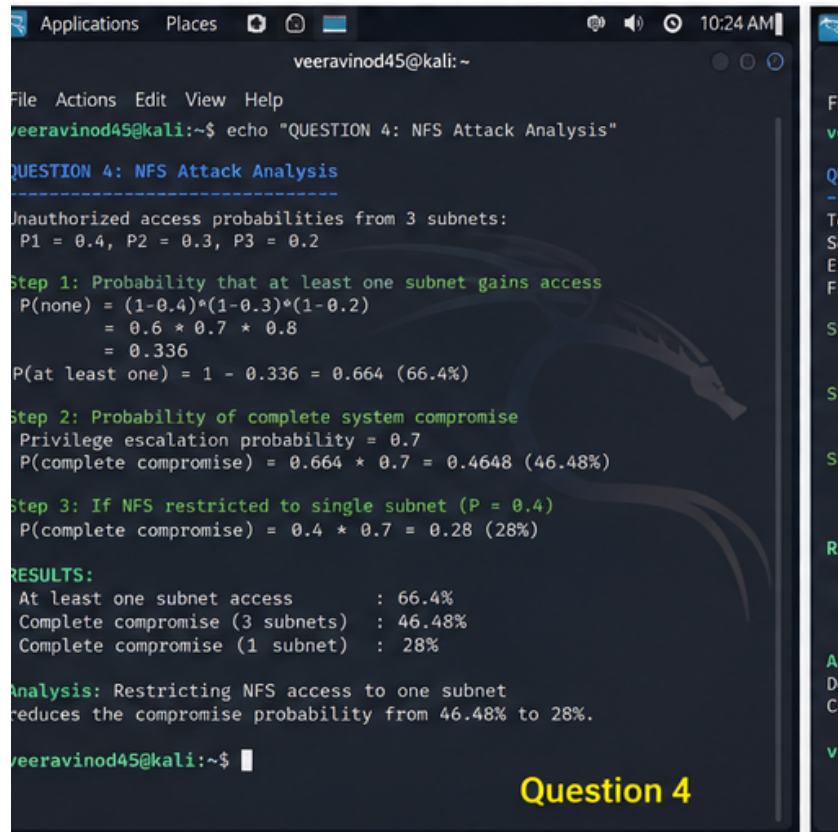
```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ echo "QUESTION 3: Cloud Linux Server Analysis"  
QUESTION 3: Cloud Linux Server Analysis  
-----  
Vulnerability probabilities of 4 services:  
P1 = 0.3, P2 = 0.2, P3 = 0.25, P4 = 0.15  
  
Step 1: Probability that at least one service is vulnerable  
P(none) = (1-0.3)*(1-0.3)*(1-0.25)*(1-0.15)  
         = 0.7 * 0.8 * 0.75 * 0.85  
         = 0.357  
P(at least one) = 1 - 0.357 = 0.643 (64.3%)  
  
Step 2: Probability of full system compromise  
Privilege escalation probability (weak permissions) = 0.5  
P(full compromise) = 0.643 * 0.5 = 0.3215 (32.15%)  
  
RESULTS:  
At least one service vulnerable : 64.3%  
Full system compromise         : 32.15%  
  
Analysis: Strengthening file permissions alone is NOT sufficient.  
Service vulnerabilities must also be addressed.  
veeravinod45@kali:~$
```

Question 3

Question 4: NFS Unauthorized Access Analysis

- Unauthorized access probabilities: 0.4, 0.3, 0.2.
- Probability none succeeds: $0.6 \times 0.7 \times 0.8 = 0.336$.
- At least one subnet gains access: $1 - 0.336 = 0.664 = 66.4\%$.
- Complete compromise: $0.664 \times 0.7 = 0.4648 = 46.48\%$.
- If restricted to the first subnet: $0.4 \times 0.7 = 28\%$.
- Restricting NFS to one trusted subnet therefore significantly reduces the modeled compromise probability.

Kali Linux Output Screenshot



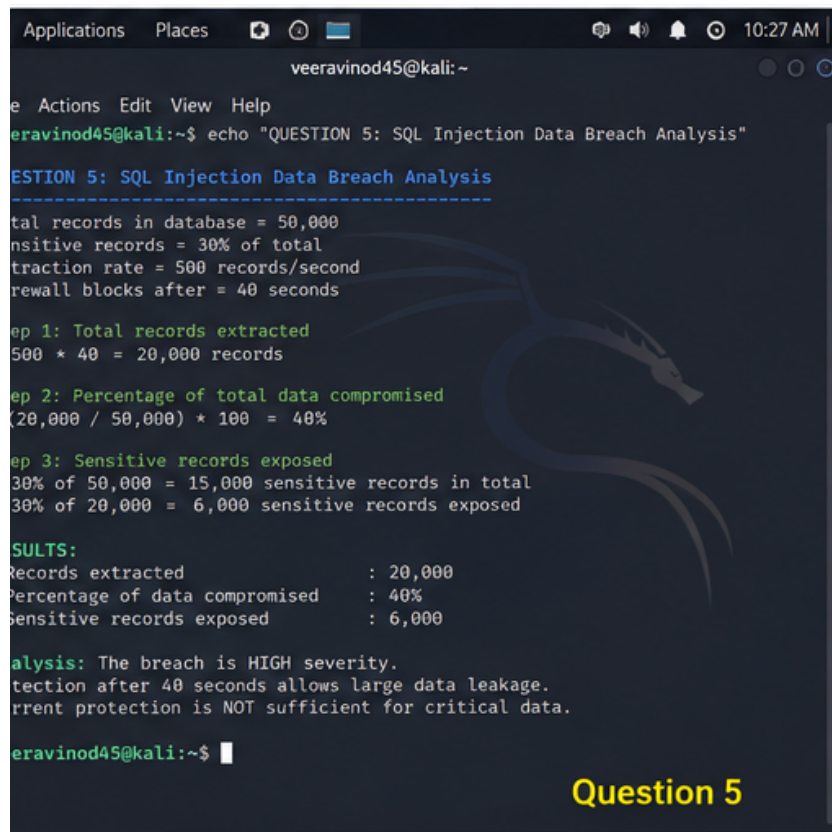
```
veeravinod45@kali:~  
File Actions Edit View Help  
veeravinod45@kali:~$ echo "QUESTION 4: NFS Attack Analysis"  
QUESTION 4: NFS Attack Analysis  
-----  
Unauthorized access probabilities from 3 subnets:  
P1 = 0.4, P2 = 0.3, P3 = 0.2  
  
Step 1: Probability that at least one subnet gains access  
P(none) = (1-0.4)*(1-0.3)*(1-0.2)  
         = 0.6 * 0.7 * 0.8  
         = 0.336  
P(at least one) = 1 - 0.336 = 0.664 (66.4%)  
  
Step 2: Probability of complete system compromise  
Privilege escalation probability = 0.7  
P(complete compromise) = 0.664 * 0.7 = 0.4648 (46.48%)  
  
Step 3: If NFS restricted to single subnet (P = 0.4)  
P(complete compromise) = 0.4 * 0.7 = 0.28 (28%)  
  
RESULTS:  
At least one subnet access      : 66.4%  
Complete compromise (3 subnets) : 46.48%  
Complete compromise (1 subnet)  : 28%  
  
Analysis: Restricting NFS access to one subnet  
reduces the compromise probability from 46.48% to 28%.  
veeravinod45@kali:~$
```

Question 4

Question 5: SQL Injection Data Breach Analysis

- Database size: 50,000 records.
- Extraction rate: 500 records/second; blocking time: 40 seconds.
- Records extracted: $500 \times 40 = 20,000$.
- Percentage compromised: $20,000 / 50,000 \times 100 = 40\%$.
- Sensitive records in the database: $50,000 \times 0.30 = 15,000$.
- Estimated sensitive records exposed: $20,000 \times 0.30 = 6,000$.
- The breach is high severity; blocking after 40 seconds still permits substantial data exfiltration.

Kali Linux Output Screenshot



```
Applications  Places  veeravinod45@kali: ~
veeravinod45@kali:~$ echo "QUESTION 5: SQL Injection Data Breach Analysis"
QUESTION 5: SQL Injection Data Breach Analysis
-----
Total records in database = 50,000
Sensitive records = 30% of total
Extraction rate = 500 records/second
Firewall blocks after = 40 seconds

Step 1: Total records extracted
500 * 40 = 20,000 records

Step 2: Percentage of total data compromised
(20,000 / 50,000) * 100 = 40%

Step 3: Sensitive records exposed
30% of 50,000 = 15,000 sensitive records in total
30% of 20,000 = 6,000 sensitive records exposed

RESULTS:
Records extracted           : 20,000
Percentage of data compromised : 40%
Sensitive records exposed    : 6,000

Analysis: The breach is HIGH severity.
Detection after 40 seconds allows large data leakage.
Current protection is NOT sufficient for critical data.

veeravinod45@kali:~$
```

Question 5